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LP-III MINI PROJECT (Group A - DAA)

Mini Project - Implement the Naive string matching algorithm and Rabin-Karp algorithm for string matching. Observe difference in working of both the algorithms for the same input.

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import timeit
def naive_search(text, pattern):
    positions = []
    n = len(text)
    m = len(pattern)

    if m == 0:
        return list(range(n + 1))

    for i in range(n - m + 1):
        if text[i:i + m] == pattern:
            positions.append(i)

    return positions



def rabin_karp(text, pattern):
    positions = []
    n = len(text)
    m = len(pattern)

    if m == 0:
        return list(range(n + 1))
```

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if m > n:
    return positions

d = 256
q = 101
h = pow(d, m - 1, q)

pattern_hash = 0
text_hash = 0

# Calculate initial hash values
for i in range(m):
    pattern_hash = (d * pattern_hash + ord(pattern[i])) % q
    text_hash = (d * text_hash + ord(text[i])) % q

# Slide the pattern over the text
for i in range(n - m + 1):
    if pattern_hash == text_hash:
        if text[i:i + m] == pattern:
            positions.append(i)

    if i < n - m:
        text_hash = (
            d * (text_hash - ord(text[i]) * h)
            + ord(text[i + m])
        ) % q

        if text_hash < 0:
            text_hash += q
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    return positions


if __name__ == "__main__":
    text = input("Enter the text: ")
    pattern = input("Enter the pattern to search: ")

    naive_result = naive_search(text, pattern)
    rabin_result = rabin_karp(text, pattern)

    print("\nNaive String Matching")
    print("Pattern found at indices:", naive_result)

    print("\nRabin-Karp Algorithm")
    print("Pattern found at indices:", rabin_result)

    repetitions = 1000

    naive_time = timeit.timeit(
        lambda: naive_search(text, pattern),
        number=repetitions
    ) / repetitions

    rabin_time = timeit.timeit(
        lambda: rabin_karp(text, pattern),
        number=repetitions
    ) / repetitions

    print("\nPerformance Comparison")
    print(f"Naive execution time: {naive_time:.8f} seconds")
    print(f"Rabin-Karp execution time: {rabin_time:.8f} seconds")
```

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OUTPUT:-

Enter the text: AABAACAADAABAABA

Enter the pattern to search: AABA

Naive String Matching

Pattern found at indices: [0, 9, 12]

Rabin-Karp Algorithm

Pattern found at indices: [0, 9, 12]

Performance Comparison

Naive execution time: 0.00000161 seconds

Rabin-Karp execution time: 0.00000400 seconds

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